Notes to write on my tables — ENGR-3017 Test 2

TJ-4510

- **p.2–3 · Joist models:** s31 and s33: 2½" flanges, 9½"–16" only · s47: 3½" flanges, all 6 depths (9½"–20"). Anything else (S37…) = no such joist. Never exterior.
- **p.6 · Top of the page:** p.6 = 40 LL / 15 DL, 5/8" and 3/4" OSB · p.7 = 40/25, 5/8" and 3/4" · p.8 = 7/8" OSB (both dead loads)
- **p.6 · Next to the 5/8" OSB block:** 5/8" OSB: no 24" o.c. → need 3/4" or 7/8" for 24". Thicker subfloor = stiffer, less vibration.
- **p.6–8 · Margin:** Bold italic = meets code but may feel bouncy. Continuous joists can use either the “simple or continuous” or the “continuous only” columns.
- **p.7 · Bottom right (if it isn't there):** Continuous: short span ÷ long span ≥ 0.4 (uplift). 14' / 6' = 0.43 ✓ · 14'-6" / 5'-6" = 0.38 ×
- **p.7 · Notes:** Minimum bearing: 1¾" at ends, 3½" at intermediate supports.
- **p.9 · Squeaks:** Squeak causes he named: joist hangers (glue / nailing), shiners (nails that missed the joist), nails that swell and back out.
- **p.10 · Detail W:** Web stiffener MAX length = depth − 3" (flanges) − ⅛" → 14" joist: 10⅞" · 11⅞": 8¾" · 9½": 6⅜". Gap at the top ⅛"–2¾". Rim options are on this page.
- **p.11 · Detail B2:** Squash blocks = joist depth + 1/16" (14" → 14-1/16"). Shorter = no.
- **p.13 · Holes:** 1½" holes OK anywhere, 3" edge-to-edge apart (or 2× the bigger hole). No holes in cantilever reinforcement. Biggest holes at mid-span = zero shear. Check Table A for the distance from the support; blank = not allowed.
- **p.14 · Cantilevers:** Backspan ≥ 2 × cantilever (4" cantilever → 8" backspan).
- **p.16 · Above the load columns:** Snow just over a column (30.20 psf) → next column UP (40). Low ≤ 6/12, High > 6/12. 18" / 20" max 3:12. Min slope ¼:12 (0.25/12). No ridge board.
- **p.19–20 · Roof details:** R9: no bird's mouth at the HIGH end. BC: bird's mouth seat can't overhang the inside face of the plate. V-cuts in shear blocking = ventilation.
- **p.21 · D factor table:** Slope length = (span × 12 + overhang) × √(1 + (pitch/12)²). D = depth × pitch ÷ 12 (his A5 key) — the table value is fine too. Order = (slope length + D) ÷ 12 → round UP to a whole foot.

TJ-9500

- **p.2–3 · Products:** LSL 1.3E & 1.55E (1.55E: 4 depths, 9½"–16") · LVL 2.0E (only one to 20") · PSL 2.0E beams (6 depths, to 19"), 1.8E columns (6 sizes). LSL ≈ OSB · LVL ≈ plywood · PSL = most voids.
- **p.5 · E values:** Lowest E (TimberStrand 1.3E) deflects the most.
- **p.6–11 · Span column:** Round the span UP to the next row (14.5' → 16.5'). Check LL, TL (* = skip), TFR. Bearing row = end / intermediate; a continuous beam over a column uses intermediate.
- **p.6–11 · Any margin:** Contractor wants it continuous? Legit only if you re-check beam → column → footing with × 1.25.
- **p.7 · Bearing:** Interior support of a continuous beam needs 2 × end bearing × 1.25 (1.9" → 4.8").
- **p.14 · Beam holes:** 1.55E LSL max hole 3" (9½") · 3⅝" (11⅞") · 4⅝" (14"–16"), at mid-depth. Spacing ≥ 2 × the bigger hole. No holes in plank orientation. LVL / PSL: no holes in the middle third of the span. Most holes → specify 1.55E LSL.
- **p.15 · Tapered end cuts:** Heel height sets the max factored reaction (5½" heel = 9,480#).
- **p.16 · Side-loaded (L17):** Find the assembly (B = 3 ply 1¾") and the nail/bolt row → plf for EITHER outside ply. 3 ply, 2 rows ½" bolts @ 24" = 585 plf.
- **p.19 · Column rows:** Wood plate rows ≠ column base rows — circle the one in the question. Microllam is NEVER a column. LSL 3½×3½ × 8' on wood plate = 8,340# (the key's "9559" is a typo for 9,595 = column base).
- **p.19 · Columns:** Steel telepost ≈ 8,000 lb (his number). TimberStrand LSL to 14', Parallam PSL to 24', L/d ≤ 50. Round height UP to the next row.
- **p.19 · Columns × 1.25:** Column × 1.25 if the beam is continuous. × 1.5625 if joists AND beam are continuous (10,000 → 12,500 → 15,625).

Method card (A5 conventions)

- **Snow:** S = 0.55 S_s + S_r (≥ 1.0 kPa) → × 20.9 → psf, **keep 2 decimals** (30.20).
- **Line loads:** psf × trib → **ROUNDUP**. TL = LL + DL. TFL = ROUNDUP(1.5 LL + 1.25 DL).
- **Trib:** Uneven spans: (Length 3 + Length 2) ÷ 2. Wing ridge RB2: Length 4 ÷ 2. Walls are 0" wide, so spans = lengths.
- **× 1.25:** Continuous joists → that floor beam's loads × 1.25. Continuous beam → its column × 1.25. Both → × 1.5625.
- **Beam span:** (Length 4 + Length 5) ÷ (interior columns + 1).
- **Stacking:** C1 = RB1 TFL × span × m · C2 = C1 + FB1 TFL × span × m · C3 = C2 + FB2 TFL × span × m. Each ROUNDUP.
- **Heights:** C1 = Length 2 × pitch (in) + Wall 1 − RB1 depth · C2 = Wall 2 − FB1 · C3 = Wall 3 − FB2 · C4 = (Length 4 ÷ 2) × pitch 2 + Wall 1 − RB2. ÷ 12, 2 decimals.
- **C4 / C5:** C4 = RB2 TFL × span ÷ 2. C5 = C4 (Floor 3 clear-spans, adds nothing). C5 height: **ask in class** (key: 10.45').
- **Order length:** Roof: (slope length + D) ÷ 12 → up. Floor simple: the span. Floor continuous: Length 3 + Length 2.
- **Footing:** psf = kPa × 20.9 · A = P ÷ psf · side = √A × 12 in. A × 1.25 if the beam goes continuous (4 → 5 ft² → 26.8").
- **Footing thickness:** Test form: (side − smallest column dimension) ÷ 2. 77.62" side, 7" column → 35.31".
- **Test form:** Column heights may be given. If nothing on p.19 carries the load at that height, write **none works** and say why.
- **Situations:** yes / no / 65-35 · document · page · detail · a reason with the number ("max span = 17'-6", 18'-0" > 17'-6"").